1. The integer n for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^{x})}{x^{n}}$ is a finite non-zero number is

NIMCET 2008

(a)1 (b)2 (c)3 (d)4

2. $\lim_{x\to\infty}\sqrt{\frac{x+\sin x}{x-\cos x}}$ is equal to

NIMCET 2011

(a)0 (b)1
(c)-1 (d)None of these

3. The value of $\lim_{n\to\infty}\frac{\pi}{n}\left[\sin\frac{\pi}{n}+\sin\frac{2\pi}{n}+\cdots+\sin\frac{(n-1)\pi}{n}\right]$ is

NIMCET 2012

(a)0 (b)π (c)2 (d)$\frac{\pi}{2}$

4. $\lim_{x\to 0}\frac{\tan x-x}{x^{2}\tan x}$ is equal to **NIMCET 2013**

(a)0 (b)1 (c)$\frac{1}{2}$ (d)$\frac{1}{3}$

5. If $f(x)=\begin{cases}\frac{\sin[x]}{[x]}, & [x]\neq 0\\ 0 & [x]=0\end{cases}$, where [x] is the largest integer but not larger than x, then $\lim_{x\to 0}f(x)$ is

NIMCET 2014

(a)-1 (b)0 (c)1 (d)Does not exist

6. The value of $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$ is

NIMCET 2015

(a)$\frac{2}{3}$ (b)$\frac{2}{\sqrt{3}}$ (c)$\frac{3\sqrt{3}}{2}$ (d)$\frac{2}{3\sqrt{3}}$

7. Evaluate $\lim_{x\to 0}\frac{x\tan x}{(1-\cos x)}$

NIMCET 2017

(a)$\frac{1}{2}$ (b)$-\frac{1}{2}$ (c)-2 (d)2

8. Let $f(x)$ be polynomial of degree four, having extreme value at $x=1$ and $x=2$. If $\lim_{x\to 0}\left[1+\frac{f(x)}{x^{2}}\right]=3$, then $f(2)$ is

NIMCET 2017

(a)0 (b)-4 (c)-8 (d)-4

9. If $\lim_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^{2}}\right)^{2x}=e^{2}$, then the value of a and b are

NIMCET 2018

(a)$a\ \epsilon\ R, b=2$ (b)$a=1, b\ \epsilon\ R$
(c)$a\ \epsilon\ R, b\ \epsilon\ R$ (d)$a=1, b=2$

10. $\lim_{x\to 3}\frac{\sqrt{3x}-3}{\sqrt{2x-4}-\sqrt{2}}$ is equal to

NIMCET 2019

(a)$\sqrt{3}$ (b)$\frac{\sqrt{3}}{2}$ (c)$\frac{1}{2\sqrt{2}}$ (d)$\frac{1}{\sqrt{2}}$

11. $\lim_{x\to 0}x^{2}e^{\sin\left(\frac{1}{x}\right)}$ is equal to

NIMCET 2020

(a)1 (b)limit does not exist
(c)infinity (d)None of these

12. $\lim_{x\to\infty}\left(\frac{x+7}{x+2}\right)^{x+5}$ equal to **NIMCET 2021**

(a)e^{5} (b)e^{-5} (c)e^{2} (d)e^{-2}

13. Which of the following is not true?

NIMCET 2022

(a) $\lim_{x\to\infty}\frac{x}{e^{x}}=0$ (b)$\lim_{x\to 0}\frac{1}{xe^{1/x}}=0$

(c) $\lim_{x\to 0^{+}}\frac{\sin x}{1+2x}=0$ (d) $\lim_{x\to 0^{+}}\frac{\cos x}{1+2x}=0$

14. If $f(x)=\lim_{x\to 0}\frac{6^{x}-3^{x}-2^{x}+1}{\log_{e}9(1-\cos x)}$ is a real number then $\lim_{x\to 0}f(x)=$

NIMCET 2023

(a)2 (b)3 (c)$\log_{e}2$ (d)$\log_{e}3$

15. $\lim_{x\to 1}\frac{x^{4}-1}{x-1}=\lim_{x\to k}\frac{x^{3}-k^{2}}{x^{2}-k^{2}}$ then find k

NIMCET 2023

(a)8/3 (b)4/3 (c)2/3 (d)1

16. Let $f(x)=\frac{(x^{2}-1)}{(|x|-1)}$. Then the value of $\lim_{x\to -1}f(x)$ is

NIMCET 2023

(a)-1 (b)1 (c)2 (d)3

17. The value of the limit (limit)

$$\lim_{x\to 0}\left(\frac{1^{x}+2^{x}+3^{x}+4^{x}}{4}\right)^{1/x} \text{ is}$$

(a) 1 (b)$3!^{1/3!}$
(c) $3!^{1/4}$ (d) $4!^{1/4}$

NIMCET 2024

18. The value of $\lim_{x\to 0}\frac{e^{x}-e^{-x}-2x}{1-\cos x}$ is equal to (Limit)

(a) 2 (b) 1
(c) 0 (d) -1

NIMCET 2024

19. What is the value of $\lim_{x\to\infty}-(x+1)\left(e^{\frac{1}{x+1}}-1\right)$?

(a) Does not exist (b) 0
(c) -1 (d) 1

NIMCET 2025

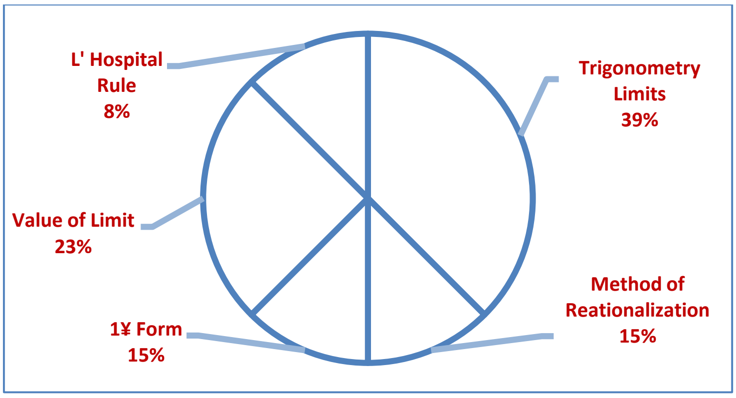
ANSWER KEY

1.	C	2.	B	3.	C	4.	D	5.	D
6.	D	7.	D	8.	A	9.	B	10.	D
11.	D	12.	A	13.	D	14.	C	15.	A
16.	c	17.	D	18.	C	19.	C		

Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	02
NIMCET 2008	01	NIMCET 2017	00
NIMCET 2009	00	NIMCET 2018	01
NIMCET 2010	00	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	01
NIMCET 2012	01	NIMCET 2021	01
NIMCET 2013	01	NIMCET 2022	01
NIMCET 2014	01	NIMCET 2023	03
NIMCET 2015	01	NIMCET 2024	02
		NIMCET 2025	01

Sub Topic Analysis



Sub Topicwise Questions Asked in NIMCET

Sub Topic	Question Asked
Trigonometry Limits	06
Method of Rationalization	02
1∞ Form	02
Value of Limit	04
L' Hospital Rule	02

SOLUTION

1. **(c)** $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ = Finite non-zero.

Using series of $\cos x$ and e^x,

$$\cos x = 1-\frac{x^2}{\lfloor} + \frac{x^4}{4} - \frac{x^6}{\lfloor 6} + \ldots\ldots$$

$$e^x = 1+\frac{x}{\lfloor 1}+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+\cdots$$

$$\lim_{x\to 0}\frac{\left[\left(1-\frac{x^2}{2}+\cdots-1\right]\right)\left[\left(1-\frac{x^2}{2}+\cdots\right)-\left(1+\frac{x}{1}+\frac{x^2}{2}+\cdots\right)\right]}{x^n}$$

$$=\lim_{x\to 0}\frac{\left[-\frac{x^2}{2}\right][-x]}{x^n}=\frac{\left(\frac{x^3}{2}\right)}{x^n}=\left(\frac{1}{2}\right)(x)^{3-n}$$

$3-n=0 \Rightarrow n=3$

2. **(b)** $\lim_{x\to\infty}\sqrt{\frac{x+\sin x}{x-\cos x}}=\lim_{x\to\infty}\sqrt{\frac{x\left(1+\frac{\sin x}{x}\right)}{x\left(1-\frac{\cos x}{x}\right)}}=1$

$\left[\because \lim_{x\to\infty}=\frac{\sin x}{x}=0 \text{ and } \lim_{x\to\infty}=\frac{\cos x}{x}=0\right]$

3. **(c)**

It is integral as a limit of sum problem.

$$=\lim_{n\to\infty}\frac{\pi}{n}\sum_{r=2}^{n}\sin\left(\frac{(r-1)\pi}{n}\right)$$

$=\lim_{n\to\infty}\frac{\pi}{n}\sum_{r=2}^{n}\sin\left(\frac{\pi r}{n}-\frac{\pi}{n}\right)$ As $n\to\infty, \frac{\pi}{n}=0$

$=\lim_{n\to\infty}\frac{\pi}{n}\sum_{r=2}^{n}\sin\left(\frac{\pi r}{n}\right)$ Put $r/n=x$ and $1/n=\mathrm{d}x$

$$=\pi\int_0^1 \sin\pi x dx=\pi\left[\frac{-\cos\pi x}{\pi}\right]_0^1=2$$

4. **(d)** $\lim_{x\to 0}\frac{\tan x-x}{x^2\tan x}\left(\frac{0}{0}\text{ form}\right)$

$$\lim_{x\to 0}\frac{\frac{\sin x}{\cos x}-x}{x^2\frac{\sin x}{\cos x}}=\lim_{x\to 0}\frac{\sin x-x\cos x}{x^2\sin x}\left(\frac{0}{0}\text{ form}\right)$$

Using L' Hospital rule

$$\lim_{x\to 0}\frac{\cos x+x\sin x-\cos x}{2x\sin x+x^2\cos x}=\lim_{x\to 0}\frac{\sin x}{2\sin x+x\cos x}\left(\frac{0}{0}\text{ form}\right)$$

Again using L' Hospital rule,

$$\lim_{x\to 0}\frac{\cos x}{2\cos x-x\sin x+\cos x}=\frac{1}{3}$$

5. **(d)**

$$f(x)=\begin{cases}\frac{\sin[x]}{[x]}, & [x]\neq 0 \Rightarrow x<0 \quad \text{or } x\geq 1\\ 0, & [x]=0, 0\leq x<1\end{cases}$$

$$\text{LHL} = \lim_{x\to 0^-}\frac{\sin[x]}{[x]}=\frac{\sin[0^-]}{[0^-]}=\frac{\sin(-1)}{-1}=\frac{-\sin(1)}{-1}$$

$=\sin(1)$

RHL $=0$

Hence, LHL ≠ RHL.

Limit does not exist

6. **(d)** $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$

$$=\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}\times\frac{\sqrt{a+2x}+\sqrt{3x}}{\sqrt{a+2x}+\sqrt{3x}}\times\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{3a+x}+2\sqrt{x}}$$

$$=\lim_{x\to a}\frac{\square+2x-3x}{3a+x-4x}\times\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}}$$

$$=\lim_{x\to a}\frac{a-x}{3(a-x)}\times\frac{2\sqrt{a}+2\sqrt{a}}{\sqrt{3a}+\sqrt{3a}}\Rightarrow\lim_{x\to a}\frac{1}{3}\cdot\frac{4\sqrt{a}}{2\sqrt{3a}}=\frac{2}{3\sqrt{3}}$$

7. **(d)** $\lim_{x\to 0}\frac{x\tan x}{(1-\cos x)}=\lim_{x\to 0}\frac{x\tan x}{1-1+2\sin^2\frac{x}{2}}$

$$=\lim_{x\to 0}\frac{x\tan x}{\frac{2\sin^2\left(\frac{x}{2}\right)}{\frac{x^2}{4}}\frac{x^2}{4}}$$

$$\lim_{x\to 0}\frac{x\tan x}{\frac{x^2}{2}}=\lim_{x\to 0}\frac{2\tan x}{x}=2$$

8. **(a)** Let $f(x)$ be a polynomial of degree four

$$f(x)=ax^4+bx^3+cx^2+dx+e$$

$$\lim_{x\to 0}\left[1+\frac{(ax^4+bx^3+cx^2+dx+e)}{x^2}\right]=3$$

$$\lim_{x\to 0}[1+ax^2+bx+c+d/x+e/x^2]=3$$

Answer of limit is finite which is 3 , hence the value of d and e must be zero.

Hence, limit is $(1+0+0+c)=3$

Hence, $c = 2$

So, $f(x)$ will be $ax^4 + bx^3 + 2x^2$.

$f(x)$ has extreme values at $x = 1$ and $x = 2$

$f'(x) = 4ax^3 + 3bx^2 + 4x$

$f'(1) = 4a + 3b + 4 = 0$

$f'(2) = 32a + 12b + 8 = 0$

Solving theses equation,

$a = 1/2, b = -2$

So,

$f(x) = (x^4/2) - 2x^3 + 2x^2$

So f(2) = 0

9. **(b)** $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}$ =e^2 [1^∞ form]

$e^{\lim\limits_{x\to\infty}\left[1+\frac{a}{x}+\frac{b}{x^2}-1\right]2x}$ = e^2

$= e^{\lim\limits_{x\to\infty}\left[\frac{a}{x}+\frac{b}{x^2}\right]2x} = e^2$

$= e^{\lim\limits_{x\to\infty}\left[a+\frac{b}{x}\right]2} = e^2$

$e^{2a} = e^2 \Rightarrow a = 1, b \in R$

10. **(d)** $\lim_{x\to 3}\frac{\sqrt{3x}-3}{\sqrt{2x-4}-\sqrt{2}}\left(\frac{0}{0}\right.$ form)

Using L' Hospital rule.

$\lim\limits_{x\to 3}\frac{\frac{1}{2\sqrt{3x}}\times 3}{\frac{1}{2\sqrt{2x-4}}\times 2} = \frac{1}{\sqrt{2}}$

11. **(d)** $\lim\limits_{x\to 0} x^2 e^{\sin\left(\frac{1}{x}\right)}$

LHL $= \lim\limits_{x\to 0^-}(0)\cdot e^{\sin\left(\frac{1}{0^-}\right)} = 0\times e^{\sin(-\infty)}$

$= 0\times e^{-1\le\sin(-\infty)\le 1} = 0$

RHL $= \lim\limits_{x\to 0^+}(0)e^{\sin\left(\frac{1}{0^+}\right)} = 0\times e^{\sin(\infty)} = 0$

LHL $=$ RHL $= 0$

12. **(a)** $\lim_{x\to\infty}\left(\frac{x+7}{x+2}\right)^{x+5} = \lim_{x\to\infty}\left(\frac{1+7/x}{1+2/x}\right)^{x+5}$ is an 1^∞ form

$\lim\limits_{x\to a^-}(f(x))^{g(x)} = e^{\lim\limits_{x\to\infty}[f(x)-1]g(x)}$

$= e^{\lim\limits_{x\to\infty}\left(\frac{x+7}{x+2}-1\right)(x+5)}$

$= e^{\lim\limits_{x\to\infty}\left(\frac{5}{x+2}\right)(x+5)}$

$= e^{5\lim\limits_{x\to\infty}\left(\frac{1+5/x}{1+2/x}\right)} = e^5$

13. **(d)** (a) $\lim_{x\to\infty}\frac{x}{e^x} = \frac{\infty}{e^\infty}\left(\frac{\infty}{\infty}\right.$ from)

Using L' Hospital rule

$\lim\limits_{x\to\infty}\frac{1}{e^x} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$

Hence, it is a right answer.

(b) $\lim_{x\to 0^+}\frac{1/x}{e^{1/x}}$

Let $\frac{1}{x} = t \to \infty = \lim_{t\to\infty}\frac{t}{e^t}$ same question as in option (a).

Hence it is also right answer.

(c) $\lim_{x\to 0^+}\frac{\sin x}{1+2x} = \lim_{x\to 0^+}\frac{(\sin x)/x}{(1/x)+2} = \frac{1}{\infty+2} = 0$

Hence, it is also right answer.

(d) $\lim_{x\to 0^+}\frac{\cos x}{1+2x} = \frac{1}{1+0} = 1.$

Hence, this is the wrong answer.

14. **(c)** $\lim_{x\to 0}\frac{(3^x-1)(2^x-1)}{2\log_e 3\left(2\sin^2\frac{x}{2}\right)}$

Using $\lim_{x\to 0}\frac{a^x-1}{x} = \log_e a$

$= \frac{1}{4\log_e 3}\lim\limits_{x\to 0}\frac{(3^x-1)}{x}\frac{(2^x-1)}{x}\frac{4\left(\frac{x^2}{4}\right)}{\sin^2\frac{x}{2}}$

$= \frac{1}{\log_e 3}\log_e 3\log_e 2 = \log_e 2$

15. **(a)** Using L' Hospital rule

$\lim\limits_{x\to 1}\frac{4x^3}{1} = \lim\limits_{x\to k}\frac{3x^2}{2x} \Rightarrow 4 = \frac{3}{2}k \Rightarrow k = \frac{8}{3}$

16. **(c)** $\lim_{x\to -1}\frac{x^2-1}{-(x+1)} = \lim_{x\to -1}\frac{(x-1)(x+1)}{-(x+1)} = 2$

17. **(d)**

$\lim\limits_{x\to 0}\left(\frac{1^x+2^x+3^x+4^x}{4}\right)^{1/x}$

$\lim\limits_{x\to 0}\left(1+\frac{1^x+2^x+3^x+4^x-1}{4}\right)^{1/x}$

$\lim\limits_{x\to 0}\left(1+\frac{1^x+2^x+3^x+4^x-4}{4}\right)^{1/x}$

$\lim\limits_{x\to 0}\left(1+\frac{1^x+2^x+3^x+4^x-4}{4}\right)$

$\lim\limits_{x\to 0}\frac{1}{4}\left(\frac{1^x-1}{x}+\frac{2^x-1}{x}+\frac{3^x-1}{x}+\frac{4^x-1}{x}\right)$

$\Rightarrow e^{1/4(\log 1\times 2\times 3\times 4)}$

= $(24)^{1/4}$ = $(4!)^{1/4}$

18. **(c)**

$\lim\limits_{x\to 0}\frac{e^x-e^{-x}-2x}{1-cosx} = \frac{0}{0}$

L Hospital Rule

$\lim\limits_{x\to 0}\frac{e^x-e^{-x}-2}{0+\sin x} = \frac{0}{0}$

Again L hospital rule

$\lim\limits_{x\to 0}\frac{e^x-e^{-x}-0}{\cos x}$

19. (c)

$\lim_{x\to\infty}(x+1)\left(e^{\frac{1}{x+1}-1}\right)$

Let $\frac{1}{x+1}=t \Rightarrow x+1=\frac{1}{t}$

Limit x → ∞ then t → 0

substitute1

$\Rightarrow \lim_{t\to 0}\frac{1}{t}(e^t-1)$

$\Rightarrow \lim_{t\to 0}-\frac{(e^t-1)}{t}$

$=\lim_{t\to 0}-\frac{\frac{d}{dt}(e^t-1)}{t}$

$=\lim_{t\to 0}-\frac{\frac{d}{dt}(e^t-1)}{\frac{d}{dt}(t)}$ (Apply 1' Hospital Rule)

$=-e^0=-1$